

Laboratory Work #2

Normalization

Task 1. Analysis of the Unnormalized Table

The initial table is:

Booking_receipt(booking_id, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, seat_numbers, ticket_price)

The table violates **First Normal Form (1NF)** because the seat_numbers column can contain multiple values in a single cell. For example, a value such as 12A, 12B contains two seat numbers in one field.

Therefore, seat_numbers is a multivalued attribute and is not atomic.

The table also contains dependencies that will need to be separated when converting the table to **2NF** and **3NF**.

Task 2. Conversion to 1NF

To convert the table to 1NF, each cell must contain only one atomic value.

Therefore, the values in seat_numbers must be separated into individual rows.

Example:

booking_id	passenger_full_name	passenger_passport_number	flight_number	seat_number	ticket_price
1	Aibek Nurgaliyev	KZ1234567	KC101	12A	150.00
1	Aruzhan Saparova	KZ2345678	KC101	12B	150.00

In the 1NF table, seat_number contains only one value in each cell.

Primary Key: (booking_id, passenger_passport_number)

The table is now in **1NF** because all attributes contain atomic values.

Task 2. Conversion to 2NF

The 1NF table has a composite primary key:

Primary Key: (booking_id, passenger_passport_number)

To convert the table to Second Normal Form (2NF), partial dependencies must be removed. A non-key attribute must depend on the whole primary key.

The following functional dependencies are identified:

- $\text{booking_id} \rightarrow \text{flight_number}, \text{ticket_price}$
- $\text{passenger_passport_number} \rightarrow \text{passenger_full_name}$

Therefore, the 1NF table contains partial dependencies. The table is decomposed into the following tables.

Booking

booking_id	flight_number	ticket_price
1	KC101	150.00
2	KC102	220.00
3	LH201	350.00
4	TK301	450.00
5	EK401	500.00

Primary Key: booking_id

Passenger

passenger_passport_number	passenger_full_name
KZ1234567	Aibek Nurgaliyev
KZ2345678	Aruzhan Saparova
KZ3456789	Dias Kassymov
KZ4567890	Madina Omarova
KZ5678901	Timur Bekov

Primary Key: passenger_passport_number

Booking_Passenger

booking_id	passenger_passport_number	seat_number
1	KZ1234567	12A
2	KZ2345678	14B
3	KZ3456789	16C
4	KZ4567890	18D
5	KZ5678901	20E

Primary Key: (booking_id, passenger_passport_number)

Foreign Keys:

- booking_id → Booking(booking_id)
- passenger_passport_number → Passenger(passenger_passport_number)

The resulting tables are in 2NF because the partial dependencies have been removed.

Task 2. Conversion to 3NF

The 2NF tables still contain transitive dependencies. To achieve Third Normal Form (3NF), non-key attributes must not depend on other non-key attributes.

The following functional dependencies are used:

- flight_number → departure_airport_name, arrival_airport_name, airline_name
- departure_airport_name → departure_city
- arrival_airport_name → arrival_city

Therefore, the flight, airport, and airline information should be separated into independent tables.

Booking

booking_id	flight_number	ticket_price
1	KC101	150.00
2	KC102	220.00
3	LH201	350.00

booking_id	flight_number	ticket_price
4	TK301	450.00
5	EK401	500.00

Primary Key: booking_id

Passenger

passenger_passport_number	passenger_full_name
KZ1234567	Aibek Nurgaliyev
KZ2345678	Aruzhan Saparova
KZ3456789	Dias Kassymov
KZ4567890	Madina Omarova
KZ5678901	Timur Bekov

Primary Key: passenger_passport_number

Booking_Passenger

booking_id	passenger_passport_number	seat_number
1	KZ1234567	12A
2	KZ2345678	14B
3	KZ3456789	16C
4	KZ4567890	18D
5	KZ5678901	20E

Primary Key: (booking_id, passenger_passport_number)

Foreign Keys:

- booking_id → Booking(booking_id)
- passenger_passport_number → Passenger(passenger_passport_number)

Flight

flight_number	departure_airport_id	arrival_airport_id	airline_id
KC101	1	2	1
KC102	2	3	2
LH201	3	4	3
TK301	4	5	4
EK401	5	1	5

Primary Key: flight_number

Foreign Keys:

- departure_airport_id → Airport(airport_id)
- arrival_airport_id → Airport(airport_id)
- airline_id → Airline(airline_id)

Airport

airport_id	airport_name	city
1	Almaty International Airport	Almaty
2	Nursultan Nazarbayev International Airport	Astana
3	Frankfurt Airport	Frankfurt
4	Istanbul Airport	Istanbul
5	Dubai International Airport	Dubai

Primary Key: airport_id

Airline

airline_id	airline_name
1	Air Astana
2	FlyArystan
3	Lufthansa

airline_id	airline_name
4	Turkish Airlines
5	Emirates

Primary Key: airline_id

The resulting database design is in **3NF** because each non-key attribute depends only on the key of its table and there are no transitive dependencies between non-key attributes.

Task 3. Functional Dependencies

The following functional dependencies were used to justify the normalization process:

Dependencies related to Booking

- booking_id → flight_number
- booking_id → ticket_price

This means that each booking determines its flight number and ticket price.

Dependencies related to Passenger

- passenger_passport_number → passenger_full_name

This means that each passport number uniquely identifies a passenger's full name.

Dependencies related to Flight

- flight_number → departure_airport_id
- flight_number → arrival_airport_id
- flight_number → airline_id

This means that each flight number determines its departure airport, arrival airport, and airline.

Dependencies related to Airport

- airport_id → airport_name
- airport_id → city

This means that each airport ID determines the airport name and city.

Dependencies related to Airline

- $\text{airline_id} \rightarrow \text{airline_name}$

This means that each airline ID determines the airline name.

Dependencies related to Booking_Passenger

- $(\text{booking_id}, \text{passenger_passport_number}) \rightarrow \text{seat_number}$

This means that the combination of booking ID and passenger passport number determines the passenger's seat number.

These functional dependencies justify the decomposition of the original unnormalized table into separate 1NF, 2NF, and 3NF tables.

Task 4. Checking the Flights Table for 3NF

The current primary key of the Flights table is:

Primary Key: flight_id

The main functional dependencies are:

- $\text{flight_id} \rightarrow \text{sch_departure_time}, \text{sch_arrival_time}$
- $\text{flight_id} \rightarrow \text{departing_airport_id}, \text{arriving_airport_id}$
- $\text{flight_id} \rightarrow \text{departing_gate}, \text{arriving_gate}$
- $\text{flight_id} \rightarrow \text{airline_id}$
- $\text{flight_id} \rightarrow \text{act_departure_time}, \text{act_arrival_time}$
- $\text{flight_id} \rightarrow \text{created_at}, \text{updated_at}$

The $\text{departing_airport_id}$, $\text{arriving_airport_id}$, and airline_id columns are foreign keys referencing other tables.

The following relationships exist:

- $\text{departing_airport_id} \rightarrow \text{Airport information}$
- $\text{arriving_airport_id} \rightarrow \text{Airport information}$
- $\text{airline_id} \rightarrow \text{Airline information}$

The Flights table does not store the airport name, city, or airline name directly. Instead, it stores only the corresponding foreign keys.

Therefore, there is no transitive dependency between non-key attributes inside the Flights table. The airport and airline information is stored in separate Airport and Airline tables.

Based on this structure, the Flights table is in **Third Normal Form (3NF)** because every non-key attribute depends on the primary key flight_id, and there are no transitive dependencies between non-key attributes.

Task 5. Implementation of the 3NF Tables

The following SQL statements implement the normalized 3NF design.

1. Airline

```
CREATE TABLE airline_3nf (  
    airline_id INT PRIMARY KEY NOT NULL,  
    airline_name VARCHAR(50) NOT NULL  
);
```

2. Airport

```
CREATE TABLE airport_3nf (  
    airport_id INT PRIMARY KEY NOT NULL,  
    airport_name VARCHAR(50) NOT NULL,  
    city VARCHAR(50) NOT NULL  
);
```

3. Passenger

```
CREATE TABLE passenger_3nf (  
    passenger_passport_number VARCHAR(20) PRIMARY KEY NOT NULL,  
    passenger_full_name VARCHAR(100) NOT NULL  
);
```

4. Flight

```
CREATE TABLE flight_3nf (  
    flight_number VARCHAR(30) PRIMARY KEY NOT NULL,  
    departure_airport_id INT NOT NULL,  
    arrival_airport_id INT NOT NULL,
```

```
airline_id INT NOT NULL,  
FOREIGN KEY (departure_airport_id) REFERENCES airport_3nf(airport_id),  
FOREIGN KEY (arrival_airport_id) REFERENCES airport_3nf(airport_id),  
FOREIGN KEY (airline_id) REFERENCES airline_3nf(airline_id)  
);
```

5. Booking

```
CREATE TABLE booking_3nf (  
    booking_id INT PRIMARY KEY NOT NULL,  
    flight_number VARCHAR(30) NOT NULL,  
    ticket_price DECIMAL(7,2) NOT NULL,  
    FOREIGN KEY (flight_number) REFERENCES flight_3nf(flight_number)  
);
```

6. Booking_Passenger

```
CREATE TABLE booking_passenger_3nf (  
    booking_id INT NOT NULL,  
    passenger_passport_number VARCHAR(20) NOT NULL,  
    seat_number VARCHAR(10) NOT NULL,  
    PRIMARY KEY (booking_id, passenger_passport_number),  
    FOREIGN KEY (booking_id) REFERENCES booking_3nf(booking_id),  
    FOREIGN KEY (passenger_passport_number) REFERENCES  
passenger_3nf(passenger_passport_number)  
);
```

Sample INSERT Statements

The following sample data is split from the original denormalized booking receipt dataset.

```
INSERT INTO airline_3nf (airline_id, airline_name) VALUES  
(1, 'Air Astana'),  
(2, 'FlyArystan'),
```

(3, 'Lufthansa'),
(4, 'Turkish Airlines'),
(5, 'Emirates');

```
INSERT INTO airport_3nf (airport_id, airport_name, city) VALUES  
(1, 'Almaty International Airport', 'Almaty'),  
(2, 'Nursultan Nazarbayev International Airport', 'Astana'),  
(3, 'Frankfurt Airport', 'Frankfurt'),  
(4, 'Istanbul Airport', 'Istanbul'),  
(5, 'Dubai International Airport', 'Dubai');
```

```
INSERT INTO passenger_3nf  
(passenger_passport_number, passenger_full_name) VALUES  
( 'KZ1234567', 'Aibek Nurgaliyev'),  
( 'KZ2345678', 'Aruzhan Saparova'),  
( 'KZ3456789', 'Dias Kassymov'),  
( 'KZ4567890', 'Madina Omarova'),  
( 'KZ5678901', 'Timur Bekov');
```

```
INSERT INTO flight_3nf  
(flight_number, departure_airport_id, arrival_airport_id, airline_id) VALUES  
( 'KC101', 1, 2, 1),  
( 'KC102', 2, 3, 2),  
( 'LH201', 3, 4, 3),  
( 'TK301', 4, 5, 4),  
( 'EK401', 5, 1, 5);
```

INSERT INTO booking_3nf

(booking_id, flight_number, ticket_price) VALUES

(1, 'KC101', 150.00),

(2, 'KC102', 220.00),

(3, 'LH201', 350.00),

(4, 'TK301', 450.00),

(5, 'EK401', 500.00);

INSERT INTO booking_passenger_3nf

(booking_id, passenger_passport_number, seat_number) VALUES

(1, 'KZ1234567', '12A'),

(2, 'KZ2345678', '14B'),

(3, 'KZ3456789', '16C'),

(4, 'KZ4567890', '18D'),

(5, 'KZ5678901', '20E');

These tables remove repeating groups, partial dependencies, and transitive dependencies from the original Booking_receipt table.

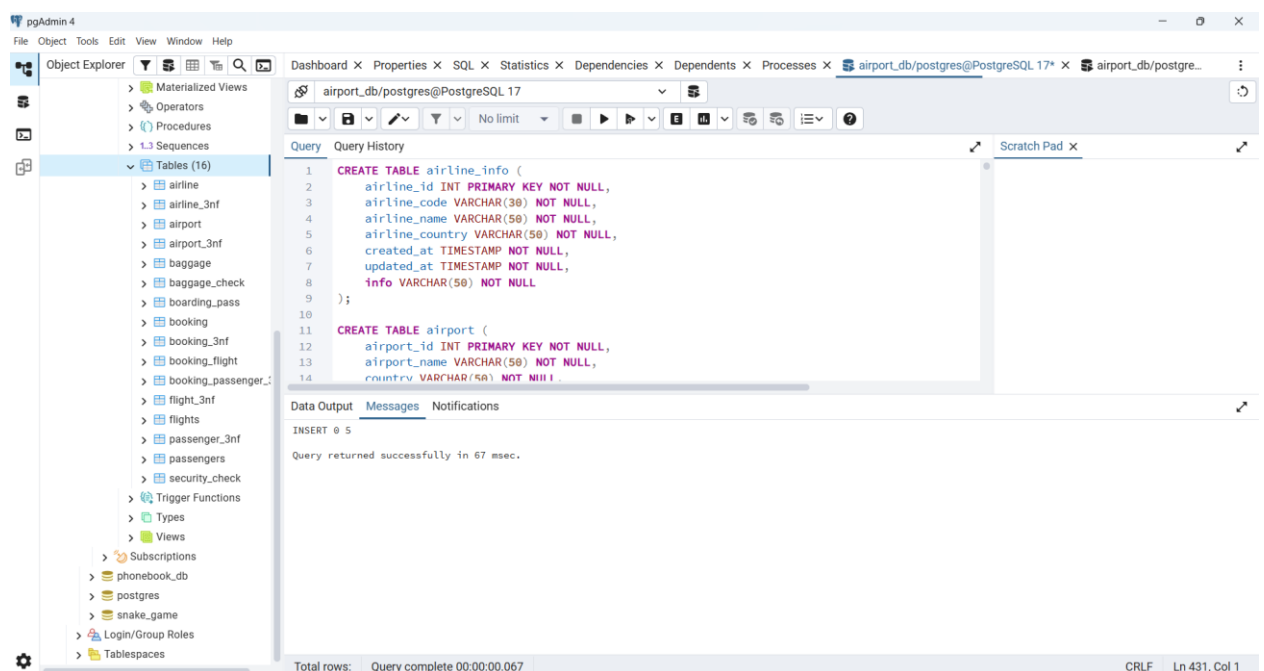


Figure 1. Database tables created in PostgreSQL



```
1 CREATE TABLE airline_info (  
2     airline_id INT PRIMARY KEY NOT NULL,  
3     airline_code VARCHAR(30) NOT NULL,  
4     airline_name VARCHAR(50) NOT NULL,  
5     airline_country VARCHAR(50) NOT NULL,  
6     created_at TIMESTAMP NOT NULL,  
7     updated_at TIMESTAMP NOT NULL,  
8     info VARCHAR(50) NOT NULL  
9 );  
10  
11 CREATE TABLE airport (  
12     airport_id INT PRIMARY KEY NOT NULL,  
13     airport_name VARCHAR(50) NOT NULL,  
14     country VARCHAR(50) NOT NULL,  
15     state VARCHAR(50) NOT NULL,  
16     city VARCHAR(50) NOT NULL,  
17     created_at TIMESTAMP NOT NULL,  
18     updated_at TIMESTAMP NOT NULL  
19 );  
20  
21 CREATE TABLE baggage_check (  
22     baggage_check_id INT PRIMARY KEY NOT NULL,  
23     check_result VARCHAR(50) NOT NULL,  
24     created_at TIMESTAMP NOT NULL
```

```
21 CREATE TABLE baggage_check (  
22     baggage_check_id INT PRIMARY KEY NOT NULL,  
23     check_result VARCHAR(50) NOT NULL,  
24     created_at TIMESTAMP NOT NULL,  
25     updated_at TIMESTAMP NOT NULL,  
26     booking_id INT NOT NULL,  
27     passenger_id INT NOT NULL  
28 );  
29  
30 CREATE TABLE baggage (  
31     baggage_id INT PRIMARY KEY NOT NULL,  
32     weight_in_kg DECIMAL(4,2) NOT NULL,  
33     created_at TIMESTAMP NOT NULL,  
34     updated_at TIMESTAMP NOT NULL,  
35     booking_id INT NOT NULL  
36 );  
37  
38 CREATE TABLE boarding_pass (  
39     boarding_pass_id INT PRIMARY KEY NOT NULL,  
40     booking_id INT NOT NULL,  
41     seat VARCHAR(50) NOT NULL,  
42     boarding_time TIMESTAMP NOT NULL,  
43     created_at TIMESTAMP NOT NULL,  
44     updated_at TIMESTAMP NOT NULL
```

```

38 CREATE TABLE boarding_pass (
39     boarding_pass_id INT PRIMARY KEY NOT NULL,
40     booking_id INT NOT NULL,
41     seat VARCHAR(50) NOT NULL,
42     boarding_time TIMESTAMP NOT NULL,
43     created_at TIMESTAMP NOT NULL,
44     updated_at TIMESTAMP NOT NULL
45 );
46
47 CREATE TABLE booking_flight (
48     booking_flight_id INT PRIMARY KEY NOT NULL,
49     booking_id INT NOT NULL,
50     flight_id INT NOT NULL,
51     created_at TIMESTAMP NOT NULL,
52     updated_at TIMESTAMP NOT NULL
53 );
54
55 CREATE TABLE booking (
56     booking_id INT PRIMARY KEY NOT NULL,
57     flight_id INT NOT NULL,
58     passenger_id INT NOT NULL,
59     booking_platform VARCHAR(50) NOT NULL,
60     created_at TIMESTAMP NOT NULL,
61     updated_at TIMESTAMP NOT NULL,

```

Query Query History

```

55 CREATE TABLE booking (
56     booking_id INT PRIMARY KEY NOT NULL,
57     flight_id INT NOT NULL,
58     passenger_id INT NOT NULL,
59     booking_platform VARCHAR(50) NOT NULL,
60     created_at TIMESTAMP NOT NULL,
61     updated_at TIMESTAMP NOT NULL,
62     status VARCHAR(50) NOT NULL,
63     price DECIMAL(7,2) NOT NULL
64 );
65
66 CREATE TABLE flights (
67     flight_id INT PRIMARY KEY NOT NULL,
68     sch_departure_time TIMESTAMP NOT NULL,
69     sch_arrival_time TIMESTAMP NOT NULL,
70     departing_airport_id INT NOT NULL,
71     arriving_airport_id INT NOT NULL,
72     departing_gate VARCHAR(50) NOT NULL,
73     arriving_gate VARCHAR(50) NOT NULL,
74     airline_id INT NOT NULL,
75     act_departure_time TIMESTAMP NOT NULL,
76     act_arrival_time TIMESTAMP NOT NULL,
77     created_at TIMESTAMP NOT NULL,
78     updated_at TIMESTAMP NOT NULL

```

```

66 CREATE TABLE flights (
67     flight_id INT PRIMARY KEY NOT NULL,
68     sch_departure_time TIMESTAMP NOT NULL,
69     sch_arrival_time TIMESTAMP NOT NULL,
70     departing_airport_id INT NOT NULL,
71     arriving_airport_id INT NOT NULL,
72     departing_gate VARCHAR(50) NOT NULL,
73     arriving_gate VARCHAR(50) NOT NULL,
74     airline_id INT NOT NULL,
75     act_departure_time TIMESTAMP NOT NULL,
76     act_arrival_time TIMESTAMP NOT NULL,
77     created_at TIMESTAMP NOT NULL,
78     updated_at TIMESTAMP NOT NULL
79 );
80
81 CREATE TABLE passengers (
82     passenger_id INT PRIMARY KEY NOT NULL,
83     first_name VARCHAR(50) NOT NULL,
84     last_name VARCHAR(50) NOT NULL,
85     date_of_birth DATE NOT NULL,
86     gender VARCHAR(50) NOT NULL,
87     country_of_citizenship VARCHAR(50) NOT NULL,
88     country_of_residence VARCHAR(50) NOT NULL,
89     passport_number VARCHAR(20) NOT NULL,
90     created_at TIMESTAMP NOT NULL

```

```

81 CREATE TABLE passengers (
82     passenger_id INT PRIMARY KEY NOT NULL,
83     first_name VARCHAR(50) NOT NULL,
84     last_name VARCHAR(50) NOT NULL,
85     date_of_birth DATE NOT NULL,
86     gender VARCHAR(50) NOT NULL,
87     country_of_citizenship VARCHAR(50) NOT NULL,
88     country_of_residence VARCHAR(50) NOT NULL,
89     passport_number VARCHAR(20) NOT NULL,
90     created_at TIMESTAMP NOT NULL,
91     updated_at TIMESTAMP NOT NULL
92 );
93
94 CREATE TABLE security_check (
95     security_check_id INT PRIMARY KEY NOT NULL,
96     check_result VARCHAR(20) NOT NULL,
97     created_at TIMESTAMP NOT NULL,
98     updated_at TIMESTAMP NOT NULL,
99     passenger_id INT NOT NULL
100 );
101
102 ALTER TABLE airline_info
103 RENAME TO airline;
104
105 ALTER TABLE booking

```

```

108 ALTER TABLE flights
109 ALTER COLUMN departing_gate TYPE TEXT;
110
111 ALTER TABLE airline
112 DROP COLUMN info;
113
114 ALTER TABLE security_check
115 ADD CONSTRAINT fk_security_passenger
116 FOREIGN KEY (passenger_id)
117 REFERENCES passengers(passenger_id);
118
119 ALTER TABLE booking
120 ADD CONSTRAINT fk_booking_passenger
121 FOREIGN KEY (passenger_id)
122 REFERENCES passengers(passenger_id);
123
124 ALTER TABLE baggage_check
125 ADD CONSTRAINT fk_baggage_check_passenger
126 FOREIGN KEY (passenger_id)
127 REFERENCES passengers(passenger_id);
128
129 ALTER TABLE baggage_check
130 ADD CONSTRAINT fk_baggage_check_booking
131 FOREIGN KEY (booking_id)
132 REFERENCES booking(booking_id);
133
134 ALTER TABLE baggage
135 ADD CONSTRAINT fk_baggage_booking
136 FOREIGN KEY (booking_id)
137 REFERENCES booking(booking_id);
138
139 ALTER TABLE boarding_pass
140 ADD CONSTRAINT fk_boarding_pass_booking
141 FOREIGN KEY (booking_id)
142 REFERENCES booking(booking_id);
143
144 ALTER TABLE booking_flight
145 ADD CONSTRAINT fk_booking_flight_booking
146 FOREIGN KEY (booking_id)
147 REFERENCES booking(booking_id);
148
149 ALTER TABLE booking_flight
150 ADD CONSTRAINT fk_booking_flight_flight
151 FOREIGN KEY (flight_id)
152 REFERENCES flights(flight_id);
153

```

```

154 ALTER TABLE flights
155 ADD CONSTRAINT fk_flights_departing_airport
156 FOREIGN KEY (departing_airport_id)
157 REFERENCES airport(airport_id);
158
159 ALTER TABLE flights
160 ADD CONSTRAINT fk_flights_arriving_airport
161 FOREIGN KEY (arriving_airport_id)
162 REFERENCES airport(airport_id);
163
164 ALTER TABLE flights
165 ADD CONSTRAINT fk_flights_airline
166 FOREIGN KEY (airline_id)
167 REFERENCES airline(airline_id);
168
169 INSERT INTO airline
170 (airline_id, airline_code, airline_name, airline_country, created_at, updated_at)
171 VALUES
172 (1, 'KC', 'Air Astana', 'Kazakhstan', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
173 (2, 'FS', 'FlyArystan', 'Kazakhstan', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
174 (3, 'LH', 'Lufthansa', 'Germany', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
175 (4, 'TK', 'Turkish Airlines', 'Turkey', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
176 (5, 'EK', 'Emirates', 'United Arab Emirates', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
177

```

```

178 INSERT INTO airport
179 (airport_id, airport_name, country, state, city, created_at, updated_at)
180 VALUES
181 (1, 'Almaty International Airport', 'Kazakhstan', 'Almaty', 'Almaty', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
182 (2, 'Nursultan Nazarbayev International Airport', 'Kazakhstan', 'Astana', 'Astana', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
183 (3, 'Frankfurt Airport', 'Germany', 'Hesse', 'Frankfurt', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
184 (4, 'Istanbul Airport', 'Turkey', 'Istanbul', 'Istanbul', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
185 (5, 'Dubai International Airport', 'United Arab Emirates', 'Dubai', 'Dubai', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
186
187
188 INSERT INTO passengers
189 (passenger_id, first_name, last_name, date_of_birth, gender,
190 country_of_citizenship, country_of_residence, passport_number,
191 created_at, updated_at)
192 VALUES
193 (1, 'Aibek', 'Nurgaliyev', '1995-03-15', 'Male',
194 'Kazakhstan', 'Kazakhstan', 'KZ1234567',
195 CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
196
197 (2, 'Aruzhan', 'Saparova', '1998-07-22', 'Female',
198 'Kazakhstan', 'Kazakhstan', 'KZ2345678',
199 CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
200
201 (3, 'Dias', 'Kassymov', '1992-11-10', 'Male',
202 'Kazakhstan', 'Germany', 'KZ3456789',
203 CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
204
205 (4, 'Madina', 'Omarova', '2000-01-30', 'Female',
206 'Kazakhstan', 'Turkey', 'KZ4567890',
207 CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
208

```

```

209      (5, 'Timur', 'Bekov', '1990-09-05', 'Male',
210        'Kazakhstan', 'Kazakhstan', 'KZ5678901',
211        CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
212
213
214  INSERT INTO flights
215  (
216    flight_id,
217    sch_departure_time,
218    sch_arrival_time,
219    departing_airport_id,
220    arriving_airport_id,
221    departing_gate,
222    arriving_gate,
223    airline_id,
224    act_departure_time,
225    act_arrival_time,
226    created_at,
227    updated_at
228  )
229  VALUES
230  (
231    1,
232    '2026-09-21 08:00:00',
233    '2026-09-21 09:30:00',
234    1,
235    2,
236    'A1',
237    'B2',
238    1,
239    '2026-09-21 08:10:00',

```

```

239    '2026-09-21 08:10:00',
240    '2026-09-21 09:35:00',
241    CURRENT_TIMESTAMP,
242    CURRENT_TIMESTAMP
243  ),
244  (
245    2,
246    '2026-09-21 10:00:00',
247    '2026-09-21 13:00:00',
248    2,
249    3,
250    'C1',
251    'D2',
252    2,
253    '2026-09-21 10:05:00',
254    '2026-09-21 13:10:00',
255    CURRENT_TIMESTAMP,
256    CURRENT_TIMESTAMP
257  ),
258  (
259    3,
260    '2026-09-22 07:30:00',
261    '2026-09-22 10:30:00',
262    3,
263    4,
264    'E1',
265    'F2',
266    3,
267    '2026-09-22 07:40:00',
268    '2026-09-22 10:35:00',
269    CURRENT_TIMESTAMP,

```

```

270     CURRENT_TIMESTAMP
271 ),
272 (
273     4,
274     '2026-09-22 14:00:00',
275     '2026-09-22 17:00:00',
276     4,
277     5,
278     'G1',
279     'H2',
280     4,
281     '2026-09-22 14:05:00',
282     '2026-09-22 17:10:00',
283     CURRENT_TIMESTAMP,
284     CURRENT_TIMESTAMP
285 ),
286 (
287     5,
288     '2026-09-23 18:00:00',
289     '2026-09-23 21:00:00',
290     5,
291     1,
292     'J1',
293     'K2',
294     5,
295     '2026-09-23 18:15:00',
296     '2026-09-23 21:10:00',
297     CURRENT_TIMESTAMP,
298     CURRENT_TIMESTAMP
299 );
300

```

```

302 INSERT INTO booking
303 (
304     booking_id,
305     flight_id,
306     passenger_id,
307     booking_platform,
308     created_at,
309     updated_at,
310     status,
311     ticket_price
312 )
313 VALUES
314 (
315     1,
316     1,
317     1,
318     'Website',
319     CURRENT_TIMESTAMP,
320     CURRENT_TIMESTAMP,
321     'Confirmed',
322     150.00
323 ),
324 (
325     2,
326     2,
327     2,
328     'Mobile App',
329     CURRENT_TIMESTAMP,
330     CURRENT_TIMESTAMP,
331     'Confirmed',
332     220.00

```

```

333     ),
334     (
335         3,
336         3,
337         3,
338         'Website',
339         CURRENT_TIMESTAMP,
340         CURRENT_TIMESTAMP,
341         'Pending',
342         350.00
343     ),
344     (
345         4,
346         4,
347         4,
348         'Travel Agency',
349         CURRENT_TIMESTAMP,
350         CURRENT_TIMESTAMP,
351         'Confirmed',
352         450.00
353     ),
354     (
355         5,
356         5,
357         5,
358         'Mobile App',
359         CURRENT_TIMESTAMP,
360         CURRENT_TIMESTAMP,
361         'Cancelled',
362         500.00
363     ):

```

```

366 INSERT INTO booking_flight
367 (
368     booking_flight_id,
369     booking_id,
370     flight_id,
371     created_at,
372     updated_at
373 )
374 VALUES
375 (1, 1, 1, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
376 (2, 2, 2, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
377 (3, 3, 3, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
378 (4, 4, 4, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
379 (5, 5, 5, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);

```

```

382 INSERT INTO baggage
383 (
384     baggage_id,
385     weight_in_kg,
386     created_at,
387     updated_at,
388     booking_id
389 )
390 VALUES
391 (1, 18.50, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 1),
392 (2, 22.30, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 2),
393 (3, 15.75, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 3),
394 (4, 25.00, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 4),
395 (5, 19.20, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 5);
396

```

```
398 INSERT INTO baggage_check
399 (
400     baggage_check_id,
401     check_result,
402     created_at,
403     updated_at,
404     booking_id,
405     passenger_id
406 )
407 VALUES
408 (1, 'Passed', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 1, 1),
409 (2, 'Passed', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 2, 2),
410 (3, 'Passed', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 3, 3),
411 (4, 'Passed', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 4, 4),
412 (5, 'Passed', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 5, 5);
413
414
415 INSERT INTO boarding_pass
416 (
417     boarding_pass_id,
418     booking_id,
419     seat,
420     boarding_time,
421     created_at,
422     updated_at
423 )
424 VALUES
425 (1, 1, '12A', '2026-09-21 07:30:00', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
426 (2, 2, '14B', '2026-09-21 09:30:00', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
427 (3, 3, '16C', '2026-09-22 07:00:00', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
428 (4, 4, '18D', '2026-09-22 13:30:00', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
```

```
419     seat,
420     boarding_time,
421     created_at,
422     updated_at
423 )
424 VALUES
425 (1, 1, '12A', '2026-09-21 07:30:00', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
426 (2, 2, '14B', '2026-09-21 09:30:00', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
427 (3, 3, '16C', '2026-09-22 07:00:00', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
428 (4, 4, '18D', '2026-09-22 13:30:00', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
429 (5, 5, '20E', '2026-09-23 17:30:00', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
430
431
432 INSERT INTO security_check
433 (
434     security_check_id,
435     check_result,
436     created_at,
437     updated_at,
438     passenger_id
439 )
440 VALUES
441 (1, 'Passed', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 1),
442 (2, 'Passed', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 2),
443 (3, 'Passed', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 3),
444 (4, 'Passed', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 4),
445 (5, 'Passed', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 5);
446
447
448
```

```

1 CREATE TABLE airline_3nf (
2     airline_id INT PRIMARY KEY NOT NULL,
3     airline_name VARCHAR(50) NOT NULL
4 );
5
6 CREATE TABLE airport_3nf (
7     airport_id INT PRIMARY KEY NOT NULL,
8     airport_name VARCHAR(50) NOT NULL,
9     city VARCHAR(50) NOT NULL
10 );
11
12 CREATE TABLE passenger_3nf (
13     passenger_passport_number VARCHAR(20) PRIMARY KEY NOT NULL,
14     passenger_full_name VARCHAR(100) NOT NULL
15 );
16
17 CREATE TABLE flight_3nf (
18     flight_number VARCHAR(30) PRIMARY KEY NOT NULL,
19     departure_airport_id INT NOT NULL,
20     arrival_airport_id INT NOT NULL,
21     airline_id INT NOT NULL,
22     FOREIGN KEY (departure_airport_id) REFERENCES airport_3nf(airport_id),
23     FOREIGN KEY (arrival_airport_id) REFERENCES airport_3nf(airport_id),
24     FOREIGN KEY (airline_id) REFERENCES airline_3nf(airline_id)
25 );
26
27 CREATE TABLE booking_3nf (
28     booking_id INT PRIMARY KEY NOT NULL,
29     flight_number VARCHAR(30) NOT NULL,
30     ticket_price DECIMAL(7,2) NOT NULL,
31     FOREIGN KEY (flight_number) REFERENCES flight_3nf(flight_number)

```

```

32 );
33
34 CREATE TABLE booking_passenger_3nf (
35     booking_id INT NOT NULL,
36     passenger_passport_number VARCHAR(20) NOT NULL,
37     seat_number VARCHAR(10) NOT NULL,
38     PRIMARY KEY (booking_id, passenger_passport_number),
39     FOREIGN KEY (booking_id) REFERENCES booking_3nf(booking_id),
40     FOREIGN KEY (passenger_passport_number)
41         REFERENCES passenger_3nf(passenger_passport_number)
42 );
43
44
45 INSERT INTO airline_3nf (airline_id, airline_name) VALUES
46 (1, 'Air Astana'),
47 (2, 'FlyArystan'),
48 (3, 'Lufthansa'),
49 (4, 'Turkish Airlines'),
50 (5, 'Emirates');
51
52 INSERT INTO airport_3nf (airport_id, airport_name, city) VALUES
53 (1, 'Almaty International Airport', 'Almaty'),
54 (2, 'Nursultan Nazarbayev International Airport', 'Astana'),
55 (3, 'Frankfurt Airport', 'Frankfurt'),
56 (4, 'Istanbul Airport', 'Istanbul'),
57 (5, 'Dubai International Airport', 'Dubai');
58
59 INSERT INTO passenger_3nf
60 (passenger_passport_number, passenger_full_name) VALUES
61 ('KZ1234567', 'Aibek Nurgaliyev'),

```

```
61 ('KZ1234567', 'Aibek Nurgaliyev'),
62 ('KZ2345678', 'Aruzhan Saparova'),
63 ('KZ3456789', 'Dias Kassymov'),
64 ('KZ4567890', 'Madina Omarova'),
65 ('KZ5678901', 'Timur Bekov');
66
67
68 INSERT INTO flight_3nf
69 (flight_number, departure_airport_id, arrival_airport_id, airline_id) VALUES
70 ('KC101', 1, 2, 1),
71 ('KC102', 2, 3, 2),
72 ('LH201', 3, 4, 3),
73 ('TK301', 4, 5, 4),
74 ('EK401', 5, 1, 5);
75
76 INSERT INTO booking_3nf
77 (booking_id, flight_number, ticket_price) VALUES
78 (1, 'KC101', 150.00),
79 (2, 'KC102', 220.00),
80 (3, 'LH201', 350.00),
81 (4, 'TK301', 450.00),
82 (5, 'EK401', 500.00);
83
84 INSERT INTO booking_passenger_3nf
85 (booking_id, passenger_passport_number, seat_number) VALUES
86 (1, 'KZ1234567', '12A'),
87 (2, 'KZ2345678', '14B'),
88 (3, 'KZ3456789', '16C'),
89 (4, 'KZ4567890', '18D'),
90 (5, 'KZ5678901', '20E');
91
```